

day-44-mcp-tool-discovery

Day 44 â MCP Tool Discovery

Date: 2026-05-09

Status: PARTIAL PASS

Blocker class: BLOCKED

Exact blocker: 'mcp_server_not_configured'

Lane

MCP per-server tool/schema discovery.

Objective

Ensure MCP tool discovery is real when a server is configured and reachable, truthful when unavailable, and never presents zero-tool or unavailable discovery as connected/live. Tool invocation and writes must remain disabled.

Inventory

Reviewed active tool discovery surfaces:

- 'GET /api/mcp/servers/:id/tools'
- 'src/lib/mcp-server-tool-schemas.ts'
- 'src/lib/zapier-tool-bridge.ts'
- Agent Zero ecosystem context MCP tool summaries
- Gateway / connector readiness references to MCP tools
- MCP button contracts for Refresh, Test server, and Enable/Disable/Reauth

Existing protected behavior retained:

- Tool discovery route is authenticated.
- Tools/list is read-only.
- No MCP tool is invoked.
- Mutating MCP config actions require owner approval.

Implemented

Added canonical owner-facing status fields to MCP tool discovery results:

- 'canonical_status'
- 'blocker_class'
- 'owner_status'
- 'read_enabled'
- 'bridge_session_required:true'
- 'approval_required_for_writes:true'
- 'allowed_owner_statuses'
- 'read_tools_total'
- 'write_tools_total'
- 'unknown_tools_total'

Tool discovery now classifies:

- Live read-only schemas as 'READY'.
- Missing OAuth/auth as 'CREDENTIAL_GATED'.
- Failed tools/list calls as 'SERVICE_DOWN'.
- Missing or unsupported server config as 'BLOCKED'.

Every discovered tool still includes:

- 'write_classification'
- 'approval_required'
- 'execution_enabled:false'

- 'writes_enabled:false'
- schema availability and required fields

Files Changed

- 'src/lib/mcp-server-tool-schemas.ts'
- 'src/lib/mcp-server-tool-schemas.test.ts'

Routes Changed

- 'GET /api/mcp/servers/:id/tools'

No execution route was added.

UI / Owner-Facing Behavior

Downstream Mission Control/Gateway/Agent Hub/connector surfaces can now distinguish:

- Tool schemas visible and read-only.
- Tool discovery blocked because the server is not configured.
- Tool discovery credential-gated.
- Tool discovery service-down.

This prevents a zero-tool or unavailable MCP tools response from being shown as connected/live.

Runtime / Service Behavior

Mission Control was rebuilt and restarted after the source change.

- Source commit: '827221a'
- Runtime PID after restart: '81357'
- Runtime bind: '127.0.0.1:3337'
- Public exposure: none; listener remains local-only.

Runtime tool discovery proof:

- Unauthenticated 'GET /api/mcp/servers/sample/tools': 401
- Authenticated 'GET /api/mcp/servers/sample/tools': 503
- Tool route canonical status: 'BLOCKED'
- Tool route blocker class: 'BLOCKED'
- Tool route error: 'mcp_server_not_configured'
- Tool count: 0
- Read tools: 0
- Write tools: 0
- Unknown tools: 0
- No tool invocation: true
- Execution enabled: false
- Writes enabled: false

Registry context during proof:

- Authenticated 'GET /api/mcp/servers': 200
- MCP server count: 0
- Registry canonical status: 'SERVICE_DOWN'
- Registry blocker class: 'SERVICE_DOWN'

Proof Artifact

- 'runtime/day-44-mcp-tool-discovery-proof.json'

The proof artifact contains summarized response fields only. No API key, token, auth file, raw tool response, or secret value was printed.

Validation

- 'git diff --check': PASS
- 'pnpm run typecheck': PASS
- 'pnpm run build': PASS
- 'pnpm test': PASS â 165 files / 1342 tests
- 'pnpm test src/lib/mcp-server-tool-schemas.test.ts': PASS
- 'pnpm test src/lib/mcp-server-registry.test.ts': PASS
- 'node scripts/check-protected-file-invariants.mjs': PASS
- Staged secret scan: PASS
- '.env' diff: clean

Blocker Classification

'BLOCKED'

Reason:

- The current runtime has no configured MCP server named 'sample', so per-server tool discovery correctly returns 'mcp_server_not_configured'.
- The broader MCP registry remains 'SERVICE_DOWN' because the local runtime cannot discover Claude MCP servers and no configured MCP servers are visible.

Developer-side closure is complete:

- Tool discovery result contract exists.
- Tool schemas remain read-only.
- Unavailable/zero-tool states are truthful.
- Tests and runtime proof pass.

Owner/admin action for live tool discovery:

- Configure at least one approved MCP server in Claude MCP config or '.claude.json'.
- Ensure the Mission Control runtime can discover it.
- If the server is HTTP and authenticated, ensure a non-expired approved OAuth/session path exists.
- Re-run authenticated 'GET /api/mcp/servers/<server>/tools'.

Commit / Push

Source commit:

- '827221a' â 'feat(mcp): add tool discovery status contract'

Push:

- 'origin/to-knowledge-mc': pushed

Day 44 report/proof commit:

- Recorded after this report is committed and pushed.

Rollback

Rollback source change:

```
""bash
git revert 827221a
""
```

Rollback Day 44 report/proof commit:

```
""bash
git revert <day44_report_commit_sha>
""
```

Safety Confirmation

- No '.env' changes.
- No secrets printed.
- No auth weakening.
- No public local exposure added.
- No MCP tool invocation.
- No MCP writes enabled.
- No fake connected/live tool state.
- No raw local paths in runtime proof.
- No fake GO.

Closeout Decision

Day 44 is developer-side complete with a truthful 'BLOCKED' route-level blocker for missing configured MCP server and a broader 'SERVICE_DOWN' registry blocker from Day 43. The next lane is Day 45 â MCP UI Page.

Next Day Started

Day 45 â MCP UI Page is the active next lane.